

Kinetic theory of gases

Definition

The kinetic theory of gases is a fundamental model in physics that explains the behavior of gases based on the motion of their constituent particles, typically molecules or atoms. It makes several key assumptions:

Assumptions

Gases are composed of particles: Gases are made up of tiny particles in constant random motion.

Volume occupied by particles is negligible: The volume occupied by the gas particles themselves is considered negligible compared to the volume of the container.

Particle collisions are elastic: Collisions between gas particles are perfectly elastic, meaning there is no net loss or gain of kinetic energy during collisions.

Particles are far apart: Gas particles are assumed to be very far apart in comparison to their size.

Particles have kinetic energy: Gas particles possess kinetic energy due to their motion, and temperature is related to the average kinetic energy of the particles.

Critical Constant

Critical constants are the specific parameters that characterize the behavior of a substance at its critical point. **The critical point represents the conditions at which a substance exists as both a liquid and a gas simultaneously**, and it marks the end of the liquid-gas boundary. The critical constants include critical temperature, critical pressure, and critical volume, which are essential in understanding the properties of a substance near its critical point.

Critical Temperature (T_c): **This is the highest temperature at which a substance can exist in a liquid state.** Above the critical temperature, the substance cannot condense into a liquid phase, regardless of the pressure applied. It signifies the temperature threshold for the liquid-gas phase transition.

Critical Pressure (P_c): **This is the minimum pressure needed to liquefy a gas at its critical temperature.** Beyond the critical pressure, the substance cannot exist in a gaseous state, even at temperatures above the critical temperature. It represents the pressure threshold for the gas-liquid phase transition.

Critical Volume (V_c): **This is the volume occupied by a mole of substance at its critical point.** It indicates the specific volume of the substance at the critical temperature and pressure.

Van der Waals equation

The Van der Waals equation is formulated as:

$$\left(P + a\frac{n^2}{V^2}\right)(V - nb) = nRT$$

P is the pressure of the gas.

V is the volume of the gas.

n is the number of moles of the gas.

T is the temperature.

R is the ideal gas constant.

a accounts for the attractive forces between gas molecules.

b adjusts for the volume occupied by the gas molecules themselves.

The Van der Waals equation introduces the two corrective terms, a and b , to adjust for the volume occupied by the gas molecules and the attractive forces between them. The term a corrects for the intermolecular forces of attraction, and b corrects for the volume excluded by the gas molecules themselves.

This equation provides a more accurate representation of the behavior of real gases, particularly at high pressures and low temperatures, where the ideal gas law fails to adequately describe the behavior of gases. The Van der Waals equation is valuable in understanding the deviations from ideal gas behavior and in predicting the properties of real gases under various conditions. It has significant applications in fields such as chemical engineering, physics, and thermodynamics.

Transmission of heat

Transmission of heat occurs through three primary mechanisms: conduction, convection, and radiation. Each process involves the transfer of thermal energy but operates differently based on the physical properties of the medium through which heat is transferred. Here is a brief overview of each mechanism

Conduction

Conduction is the transfer of heat through a material without any perceptible movement of the material itself. It occurs due to the collision and interaction of neighboring molecules within a substance. Materials that are good conductors, such as metals, allow heat to transfer more rapidly, while insulators, like wood or air, impede the transfer of heat. Common examples of conduction include heating a metal rod at one end and feeling the other end become warm, or touching a hot pan and feeling the heat on your hand.

Convection

Convection is the transfer of heat through the movement of fluids (liquids or gases). It involves the actual movement of the heated material from one place to another. When a fluid is heated, it becomes less dense and rises, while the cooler, denser fluid sinks. This creates a continuous circulation pattern, transferring heat throughout the medium. Convection plays a significant role in phenomena such as ocean currents, atmospheric circulation, and the heating of a room through a heating system.

Radiation

Radiation is the transfer of heat through electromagnetic waves, such as infrared radiation. Unlike conduction and convection, radiation can occur in a vacuum and does not require a medium. All objects with a temperature above absolute zero emit thermal radiation. Darker and rougher surfaces are better emitters and absorbers of heat radiation compared to lighter and smoother surfaces. Common examples of radiation include feeling the warmth of the sun or the heat radiating from a fire.

Second law of thermodynamics

Any spontaneously occurring process will always lead to an escalation in the entropy (S) of the universe.

In simple words, the law explains that an isolated system's entropy will never decrease over time.

Kelvin–Planck form of the second law of thermodynamics

It is impossible to construct a heat engine that, operating in a cycle, produces no effect other than the input of energy by heat from a reservoir and the performance of an equal amount of work.

Clausius statement

It is impossible to construct a cyclical machine whose sole effect is to transfer energy continuously by heat from one object to another object at a higher temperature without the input of energy by work.

In simpler terms, energy does not transfer spontaneously by heat from a cold object to a hot object.

Reversible and Irreversible Processes

In a reversible process, the system undergoing the process can be returned to its initial conditions along the same path on a PV diagram, and every point along this path is an equilibrium state. A process that does not satisfy these requirements is irreversible.

All natural processes are known to be irreversible.

Heat Engine

A heat engine is a device that takes in energy by heat and, operating in a cyclic process, expels a fraction of that energy by means of work.

For instance, in a typical process by which a power plant produces electricity, a fuel such as coal is burned and the high-temperature gases produced are used to convert liquid water to steam. This steam is directed at the blades of a turbine, setting it into rotation. The mechanical energy associated with this rotation is used to drive an electric generator. Another device that can be modeled as a heat engine is the internal combustion engine in an automobile. This device uses energy from a burning fuel to perform work on pistons that results in the motion of the automobile.

Basic fundamental of heat engine

A heat engine carries some working substance through a cyclic process during which

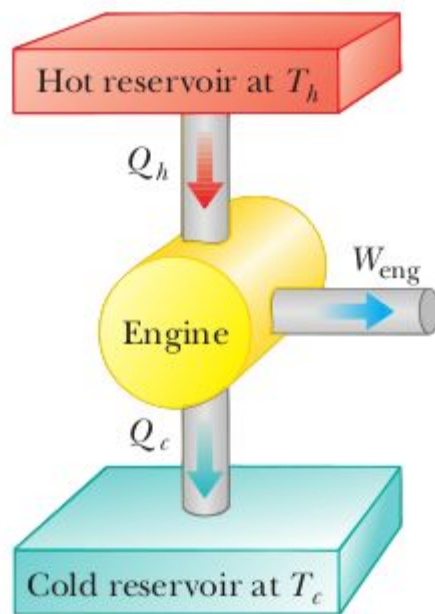
1. The working substance absorbs energy by heat from a high-temperature energy reservoir
2. Work is done by the engine
3. Energy is expelled by heat to a lower-temperature reservoir.

It is useful to represent a heat engine schematically. The engine absorbs a quantity of energy Q_h from the hot reservoir. For the mathematical discussion of heat engines, we use absolute values to make all energy transfers positive and the direction of transfer is indicated with an explicit positive or negative sign. The engine does work W_{eng} (so that negative work $W = -W_{\text{eng}}$ is done on the engine) and then gives up a quantity of energy Q_c to the cold reservoir. Because the working substance goes through a cycle, its initial and final internal energies are equal: $E_{\text{int}} = 0$. Hence, from the first law of thermodynamics,

$$E_{\text{int}} = Q + W$$

$Q - W_{\text{eng}} = 0$, and the net work W_{eng} done by a heat engine is equal to the net energy Q_{net} transferred to it.

$$Q_{\text{net}} = |Q_h| - |Q_c|$$



Thermal efficiency of a heat engine

The thermal efficiency “ e ” of a heat engine is defined as the ratio of the net work done by the engine during one cycle to the energy input at the higher temperature during the cycle:

$$e \equiv \frac{W_{\text{eng}}}{|Q_h|} = \frac{|Q_h| - |Q_c|}{|Q_h|} = 1 - \frac{|Q_c|}{|Q_h|}$$